COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31s2
DATE PERFORMED:	17/09/2026	**DATE SUBMITTED:**	19/09/2026
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ACTIVITY NO. 8 **Install, Configure, and Manage Availability Monitoring tools**

1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

GitHub new folder:
https://github.com/DamienJakeGestiada/CPE212_JakeGestiada/tree/main/HOA8

TASK 1: Create a Playbook That Installs Nagios in Both Ubuntu and CentOS

```
jakeworkstation@WS:~/JakeGestiada/HOA8$ tree
.
├── ansible.cfg
├── install_nagios.yml
├── inventory.yaml
└── roles
    └── nagios
        └── tasks
            ├── centos.yml
            ├── main.yml
            └── ubuntu.yml

4 directories, 6 files
jakeworkstation@WS:~/JakeGestiada/HOA8$ nano inventory.yaml
jakeworkstation@WS:~/JakeGestiada/HOA8$ cat inventory.yaml
---
all:
  children:
    ubuntu:
      hosts:
        server1:
          ansible_host: 192.168.56.105
          ansible_user: jakeserver

    centos:
      hosts:
        centos1:
          ansible_host: 192.168.56.104
          ansible_user: jakecentos
jakeworkstation@WS:~/JakeGestiada/HOA8$
```

```
jakeworkstation@WS:~/JakeGestiada/HOA8$ ansible all -i inventory.yaml -m ansible.builtin.ping
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
jakeworkstation@WS:~/JakeGestiada/HOA8$
```

- I created the HOA8 Nagios project structure, configured the inventory for Server1 and CentOS with their IP addresses and usernames, and tested the Ansible connection to confirm that both managed nodes responded successfully with pong.

```
 GNU nano 7.2                                  ansible.cfg
[defaults]
inventory = ./inventory.yaml
host_key_checking = False
```

- configured ansible.cfg to use the HOA8 inventory file and disabled host key checking so Ansible can connect to the managed nodes without repeated SSH host-key prompts.

```
jakeworkstation@WS:~/JakeGestiada/HOAB$ cat roles/nagios_centos/tasks/main.yml
---
- name: Update CentOS package repository
  ansible.builtin.dnf:
    update_cache: true

- name: Install Nagios Core prerequisites
  ansible.builtin.dnf:
    name:
      - gcc
      - glibc
      - glibc-common
      - perl
      - httpd
      - php
      - wget
      - gd
      - gd-devel
      - s-nail
      - postfix
      - openssl-devel
      - make
      - gettext
      - automake
      - autoconf
      - net-snmp
      - net-snmp-utils
      - epel-release
      - httpd-tools
    state: present

- name: Download Nagios Core source
  ansible.builtin.shell: >
    wget -O /tmp/nagioscore.tar.gz
    $(wget -q -O -
    https://api.github.com/repos/NagiosEnterprises/nagioscore/releases/latest |
    grep '"browser_download_url":' |
    grep -o 'https://[^"]*')
  args:
    creates: /tmp/nagioscore.tar.gz

- name: Extract Nagios Core source
  ansible.builtin.unarchive:
    src: /tmp/nagioscore.tar.gz
```

```
jakeworkstation@WS:~/JakeGestiada/HOAB$ cat nagios.yml
---
- name: Install Nagios on Ubuntu Server1
  hosts: ubuntu
  become: true
  roles:
    - nagios_ubuntu

- name: Install Nagios on CentOS
  hosts: centos
  become: true
  roles:
    - nagios_centos
jakeworkstation@WS:~/JakeGestiada/HOAB$
```

- Created the Ansible role and playbook files for Nagios, including separate configurations for

Ubuntu and CentOS. The files define the required packages, Nagios source installation, and the hosts where each role will be applied.

jakeworkstation@WS: ~/JakeGestiada/HOA8

GNU nano 7.2 roles/nagios/tasks/main.yml *

```
---
- name: Run Ubuntu Nagios tasks
  ansible.builtin.include_tasks: ubuntu.yml
  when: ansible_facts.os_family == "Debian"

- name: Run CentOS Nagios tasks
  ansible.builtin.include_tasks: centos.yml
  when: ansible_facts.os_family == "RedHat"
```

jakeworkstation@WS: ~/JakeGestiada/HOA8

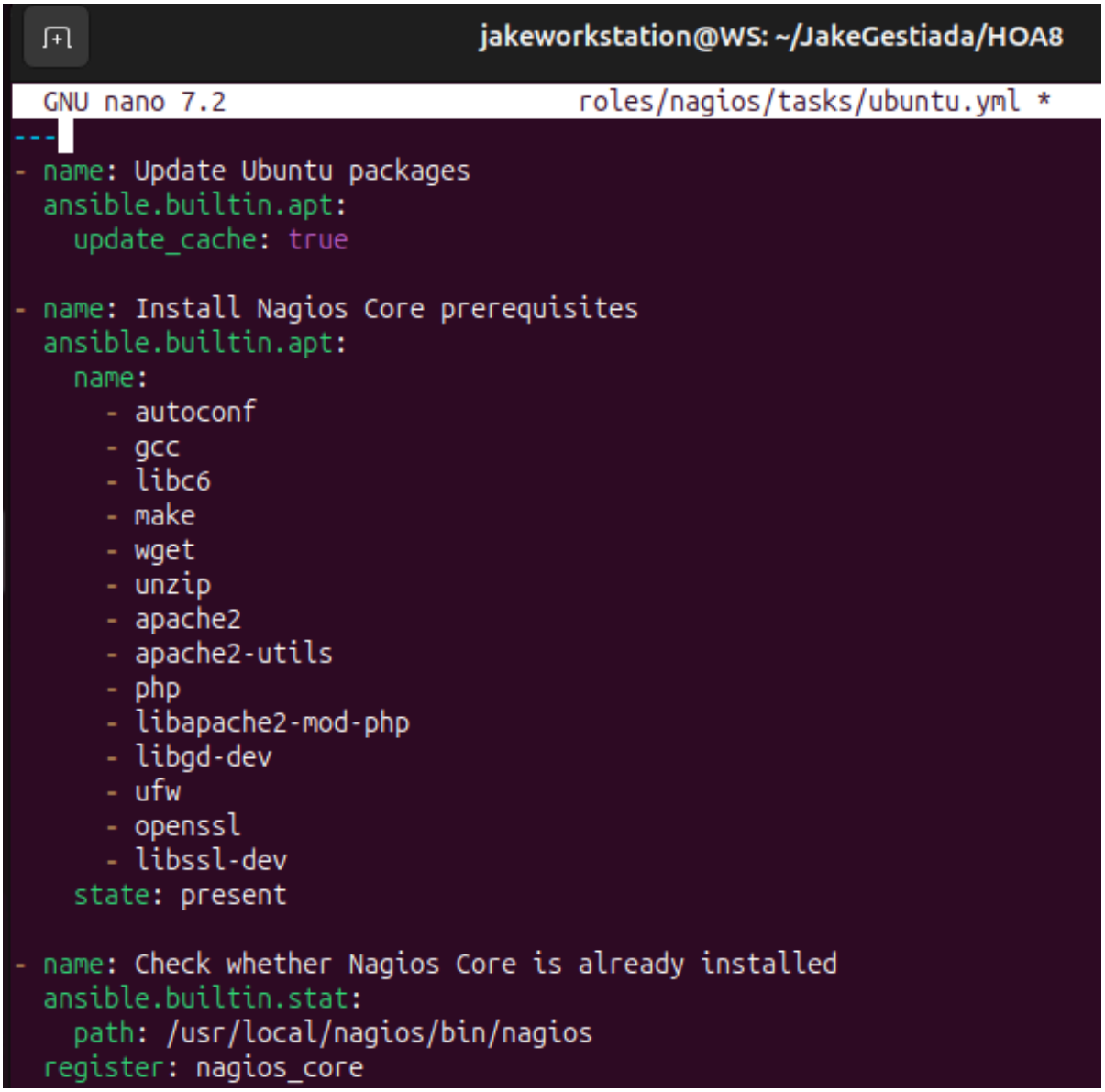
GNU nano 7.2 roles/nagios/tasks/ubuntu.yml *

```
  ansible.builtin.command:
    cmd: ufw allow Apache

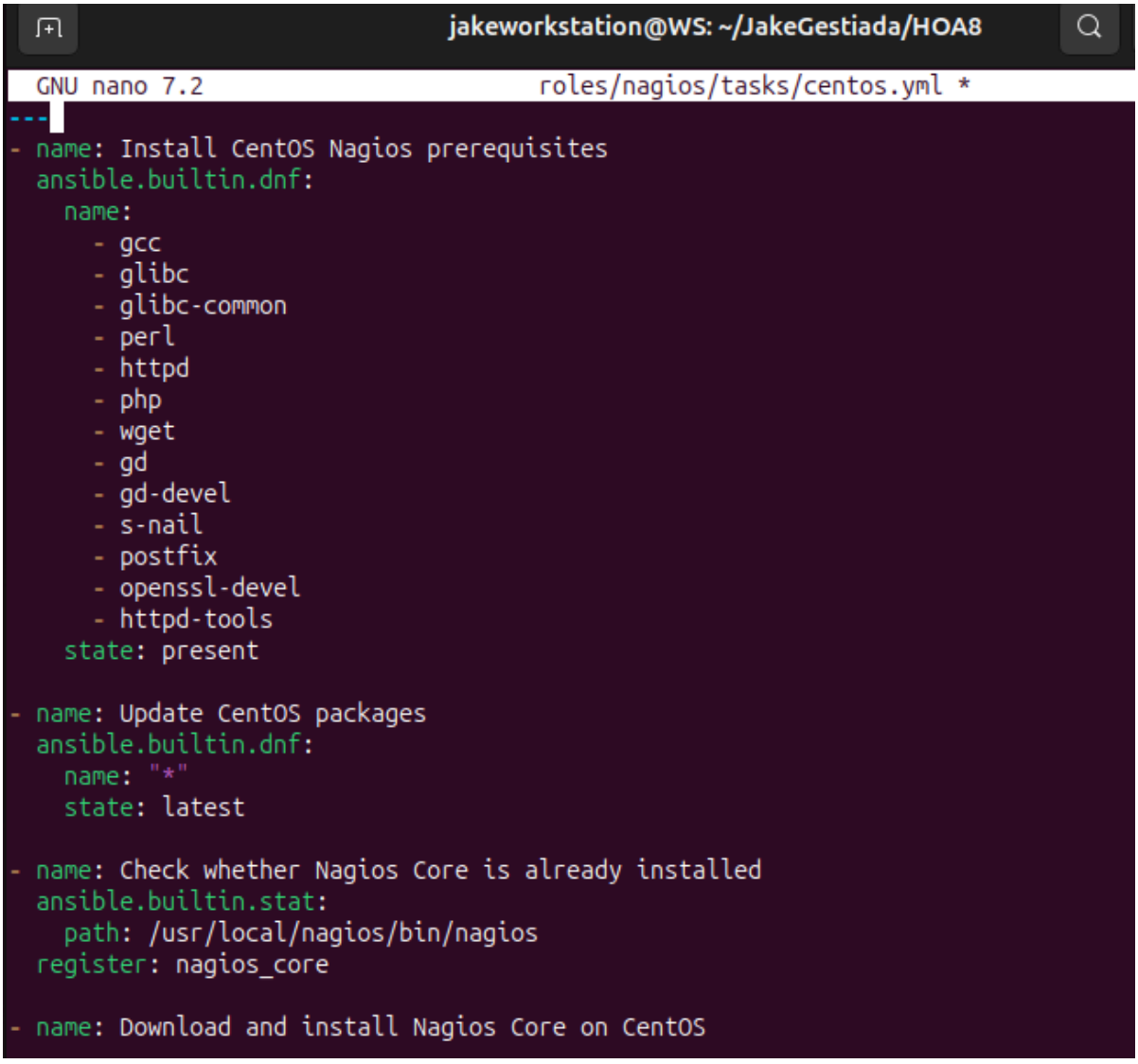
- name: Enable Apache
  ansible.builtin.service:
    name: apache2
    enabled: true
    state: restarted

- name: Enable and start Nagios
  ansible.builtin.service:
    name: nagios
    enabled: true
    state: started
```

- The role was set to choose the proper tasks for Ubuntu or CentOS automatically. The Ubuntu tasks also configure Apache and make sure Nagios starts and runs properly.



- The Ubuntu task file was configured to update the system and install the packages needed for Nagios Core. It also checks whether Nagios is already installed before continuing with the installation.



- The CentOS task file was configured with the required packages and system updates for Nagios Core. It also checks whether Nagios is already installed before starting the download and installation process.

```
jakeworkstation@WS: ~/JakeGestiada/HOA8
GNU nano 7.2                                  install_nagios.yml
---
- name: Install Nagios on Ubuntu Server1
  hosts: ubuntu
  become: true
  roles:
    - nagios

- name: Install Nagios on CentOS
  hosts: centos
  become: true
  roles:
    - nagios
```

- The main playbook was configured to apply the Nagios role to both the Ubuntu Server1 and CentOS managed nodes. Each host group uses become: true so the required installation tasks can run with administrative privileges.

```
jakeworkstation@WS:~/JakeGestiada/HOA8$ ansible-playbook install_nagios.yml -k -K
SSH password:
BECOME password[defaults to SSH password]:

PLAY [Install Nagios on Ubuntu Server1] ****************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [server1]

TASK [nagios : Run Ubuntu Nagios tasks] ****************************************************************
included: /home/jakeworkstation/JakeGestiada/HOA8/roles/nagios/tasks/ubuntu.yml for server1

TASK [nagios : Update Ubuntu packages] *****************************************************************
changed: [server1]

TASK [nagios : Install Nagios Core prerequisites] ******************************************************
ok: [server1]

TASK [nagios : Check whether Nagios Core is already installed] *****************************************
ok: [server1]

TASK [nagios : Download Nagios Core source] ************************************************************
skipping: [server1]

TASK [nagios : Extract Nagios Core source] *************************************************************
skipping: [server1]

TASK [nagios : Configure Nagios Core] ******************************************************************
```

```
TASK [nagios : Install Nagios daemon] ******************************************************************
skipping: [centos1]

TASK [nagios : Install Nagios command mode] ************************************************************
skipping: [centos1]

TASK [nagios : Install Nagios configuration] ***********************************************************
skipping: [centos1]

TASK [nagios : Install Nagios Apache configuration] ****************************************************
skipping: [centos1]

TASK [nagios : Allow HTTP through firewalld] ***********************************************************
changed: [centos1]

TASK [nagios : Reload firewalld] ***********************************************************************
changed: [centos1]

TASK [nagios : Enable and start Apache] ****************************************************************
ok: [centos1]

TASK [nagios : Enable and start Nagios] ****************************************************************
ok: [centos1]

PLAY RECAP *********************************************************************************************
centos1                    : ok=8    changed=2    unreachable=0    failed=0    skipped=12   rescued=0    ignore
d=0
server1                    : ok=11   changed=5    unreachable=0    failed=0    skipped=11   rescued=0    ignore
d=0
```

- The install nagios.yml playbook was executed on both Server1 and CentOS, and the tasks completed successfully. The final play recap shows unreachable=0 and failed=0 for both managed nodes, confirming that the Nagios deployment ran without errors.

```
jakeworkstation@WS:~/JakeGestiada/HOA8$ ansible-playbook install_nagios.yml --limit server1 -k -K
SSH password:
BECOME password[defaults to SSH password]:

PLAY [Install Nagios on Ubuntu Server1] ****************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [server1]

TASK [nagios : Run Ubuntu Nagios tasks] ****************************************************************
included: /home/jakeworkstation/JakeGestiada/HOA8/roles/nagios/tasks/ubuntu.yml for server1

TASK [nagios : Update Ubuntu packages] *****************************************************************
changed: [server1]

TASK [nagios : Install Nagios Core prerequisites] ******************************************************
ok: [server1]

TASK [nagios : Check whether Nagios Core is already installed] *****************************************
ok: [server1]

TASK [nagios : Download Nagios Core source] ************************************************************
skipping: [server1]

TASK [nagios : Extract Nagios Core source] *************************************************************
skipping: [server1]

TASK [nagios : Configure Nagios Core] ******************************************************************
skipping: [server1]
TASK [nagios : Install Nagios configuration] ***********************************************************
skipping: [server1]

TASK [nagios : Install Nagios Apache configuration] ****************************************************
skipping: [server1]

TASK [nagios : Enable Apache rewrite module] ***********************************************************
changed: [server1]

TASK [nagios : Enable Apache CGI module] ***************************************************************
changed: [server1]

TASK [nagios : Allow Apache through UFW] ***************************************************************
changed: [server1]

TASK [nagios : Enable and restart Apache] **************************************************************
changed: [server1]

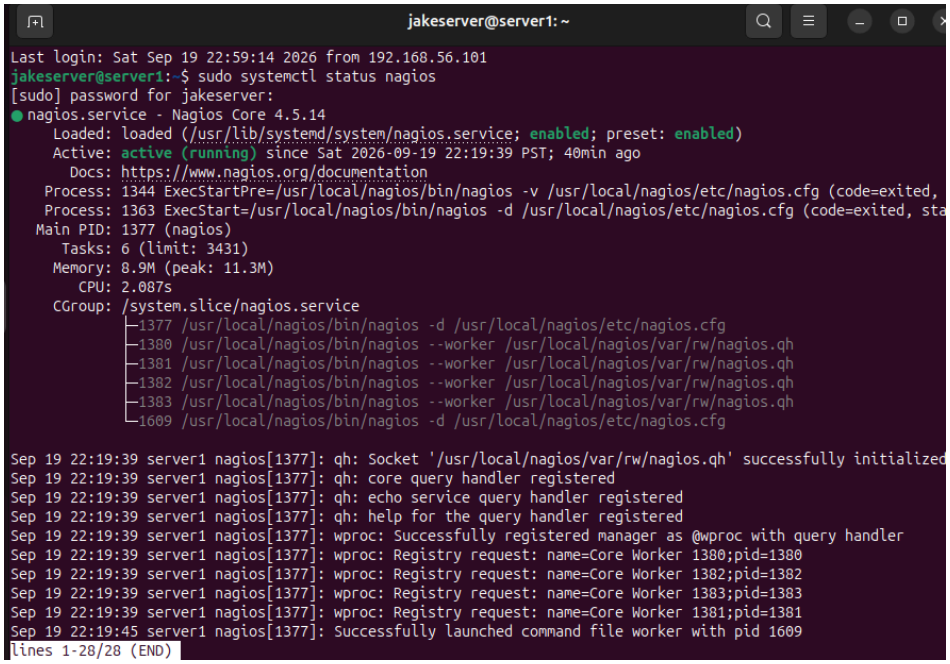
TASK [nagios : Enable and start Nagios] ****************************************************************
ok: [server1]

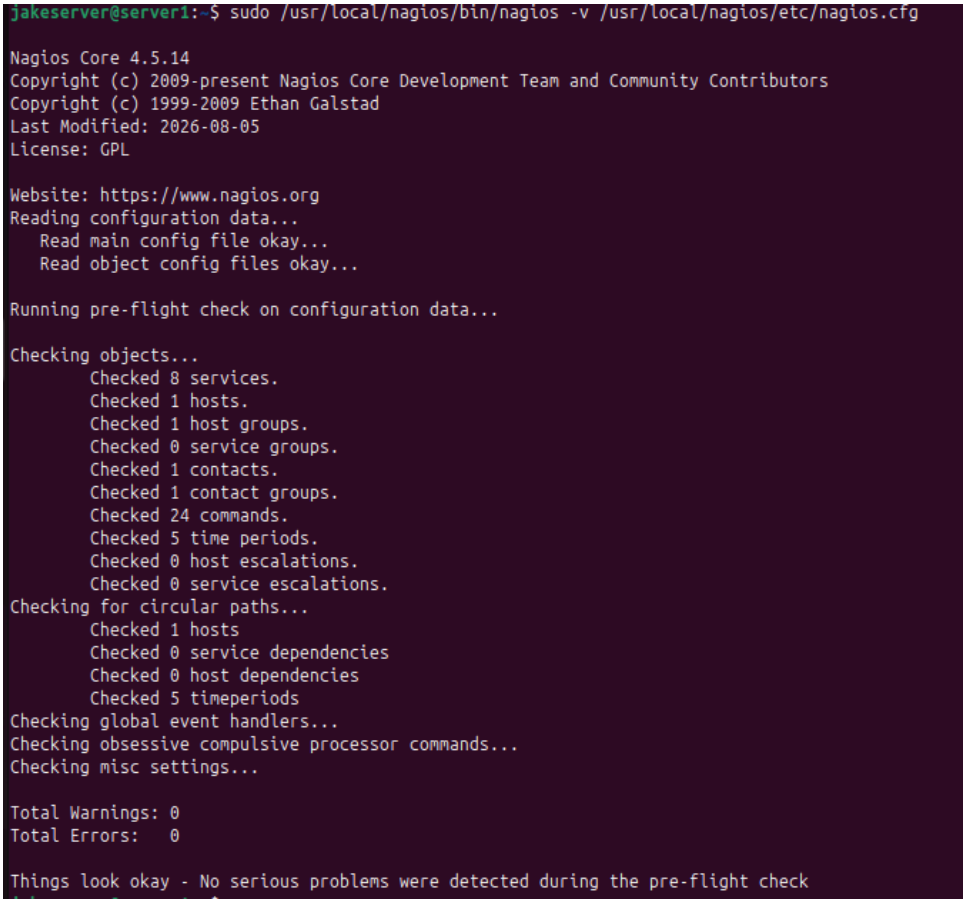
TASK [nagios : Run CentOS Nagios tasks] ****************************************************************
skipping: [server1]

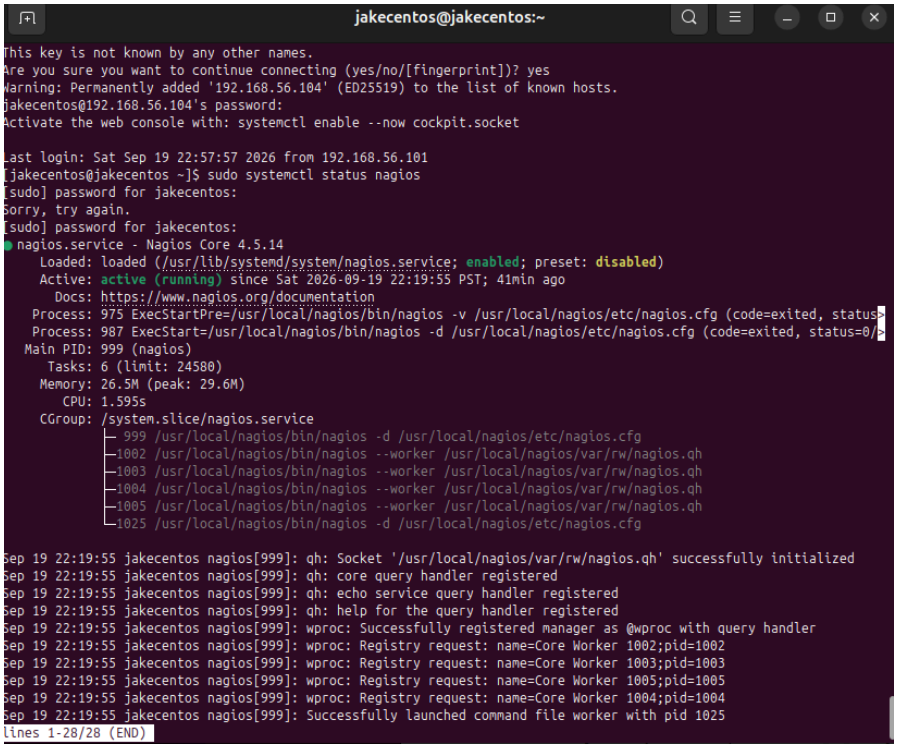
PLAY [Install Nagios on CentOS] ************************************************************************
skipping: no hosts matched

PLAY RECAP *********************************************************************************************
server1                    : ok=11   changed=5    unreachable=0    failed=0    skipped=11   rescued=0    ignore
d=0
```

- The playbook was tested on Server1 only using the --limit server1 option. The Ubuntu Nagios tasks completed successfully, while the CentOS tasks were skipped because only Server1 was selected for this run.


jakeserver@server1: ~
Last login: Sat Sep 19 22:59:14 2026 from 192.168.56.101
jakeserver@server1:~$ sudo systemctl status nagios
[sudo] password for jakeserver:
● nagios.service - Nagios Core 4.5.14
Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
Active: active (running) since Sat 2026-09-19 22:19:39 PST; 40min ago
Docs: https://www.nagios.org/documentation
Process: 1344 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg (code=exited,
Process: 1363 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg (code=exited, sta
Main PID: 1377 (nagios)
Tasks: 6 (limit: 3431)
Memory: 8.9M (peak: 11.3M)
CPU: 2.087s
CGroup: /system.slice/nagios.service
├─1377 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
├─1380 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
├─1381 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
├─1382 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
├─1383 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
└─1609 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
Sep 19 22:19:39 server1 nagios[1377]: qh: Socket '/usr/local/nagios/var/rw/nagios.qh' successfully initialized
Sep 19 22:19:39 server1 nagios[1377]: qh: core query handler registered
Sep 19 22:19:39 server1 nagios[1377]: qh: echo service query handler registered
Sep 19 22:19:39 server1 nagios[1377]: qh: help for the query handler registered
Sep 19 22:19:39 server1 nagios[1377]: wproc: Successfully registered manager as @wproc with query handler
Sep 19 22:19:39 server1 nagios[1377]: wproc: Registry request: name=Core Worker 1380;pid=1380
Sep 19 22:19:39 server1 nagios[1377]: wproc: Registry request: name=Core Worker 1382;pid=1382
Sep 19 22:19:39 server1 nagios[1377]: wproc: Registry request: name=Core Worker 1383;pid=1383
Sep 19 22:19:39 server1 nagios[1377]: wproc: Registry request: name=Core Worker 1381;pid=1381
Sep 19 22:19:45 server1 nagios[1377]: Successfully launched command file worker with pid 1609
lines 1-28/28 (END)


jakeserver@server1:~$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
Nagios Core 4.5.14
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2026-08-05
License: GPL
Website: https://www.nagios.org
Reading configuration data...
Read main config file okay...
Read object config files okay...
Running pre-flight check on configuration data...
Checking objects...
Checked 8 services.
Checked 1 hosts.
Checked 1 host groups.
Checked 0 service groups.
Checked 1 contacts.
Checked 1 contact groups.
Checked 24 commands.
Checked 5 time periods.
Checked 0 host escalations.
Checked 0 service escalations.
Checking for circular paths...
Checked 1 hosts
Checked 0 service dependencies
Checked 0 host dependencies
Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...
Total Warnings: 0
Total Errors: 0
Things look okay - No serious problems were detected during the pre-flight check




```
[jakecentos@jakecentos ~]$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.5.14
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2026-08-05
License: GPL

Website: https://www.nagios.org
Reading configuration data...
   Read main config file okay...
   Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
        Checked 8 services.
        Checked 1 hosts.
        Checked 1 host groups.
        Checked 0 service groups.
        Checked 1 contacts.
        Checked 1 contact groups.
        Checked 24 commands.
        Checked 5 time periods.
        Checked 0 host escalations.
        Checked 0 service escalations.
Checking for circular paths...
        Checked 1 hosts
        Checked 0 service dependencies
        Checked 0 host dependencies
        Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 0
Total Errors:   0

Things look okay - No serious problems were detected during the pre-flight check
[jakecentos@jakecentos ~]$
```

```
[jakecentos@jakecentos ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: active (running) since Sat 2026-09-19 22:19:55 PST; 42min ago
       Docs: man:httpd.service(8)
   Main PID: 970 (httpd)
     Status: "Total requests: 9; Idle/Busy workers 100/0;Requests/sec: 0.00354; Bytes served/sec:   2 B/sec"
      Tasks: 177 (limit: 24580)
     Memory: 20.0M (peak: 20.7M)
        CPU: 3.325s
     CGroup: /system.slice/httpd.service
             ├─ 970 /usr/sbin/httpd -DFOREGROUND
             ├─1084 /usr/sbin/httpd -DFOREGROUND
             ├─1088 /usr/sbin/httpd -DFOREGROUND
             ├─1091 /usr/sbin/httpd -DFOREGROUND
             └─1092 /usr/sbin/httpd -DFOREGROUND

Sep 19 22:19:55 jakecentos systemd[1]: Starting The Apache HTTP Server...
Sep 19 22:19:55 jakecentos httpd[970]: AH00558: httpd: Could not reliably determine the server's fully qualified do>
Sep 19 22:19:55 jakecentos httpd[970]: Server configured, listening on: port 80
Sep 19 22:19:55 jakecentos systemd[1]: Started The Apache HTTP Server.
lines 1-22/22 (END)
```

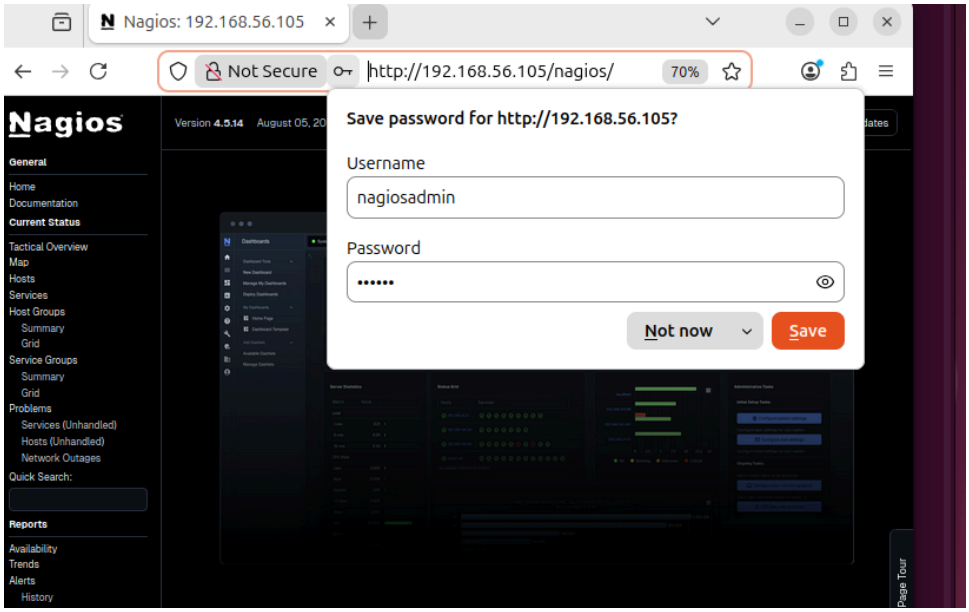
- Nagios was running properly on both Server1 and CentOS, with the services active and the configuration checks showing 0 warnings and 0 errors. This confirms that the Nagios installation and configuration were completed successfully on both managed nodes.

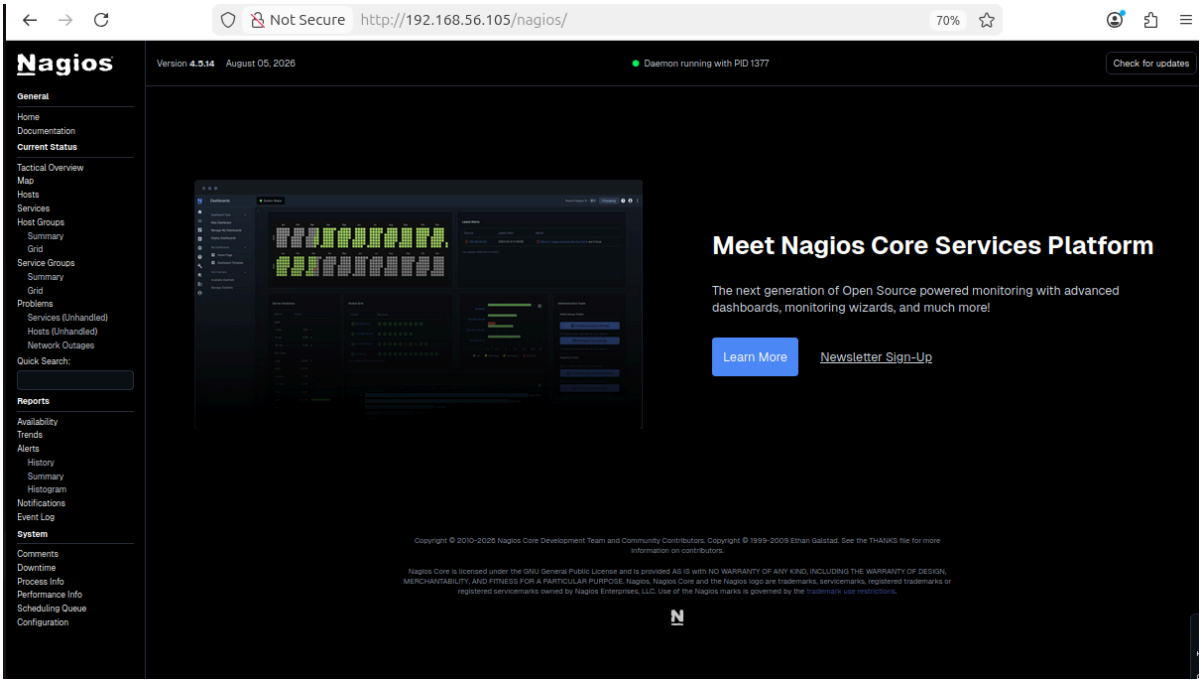
TASK 2: verification

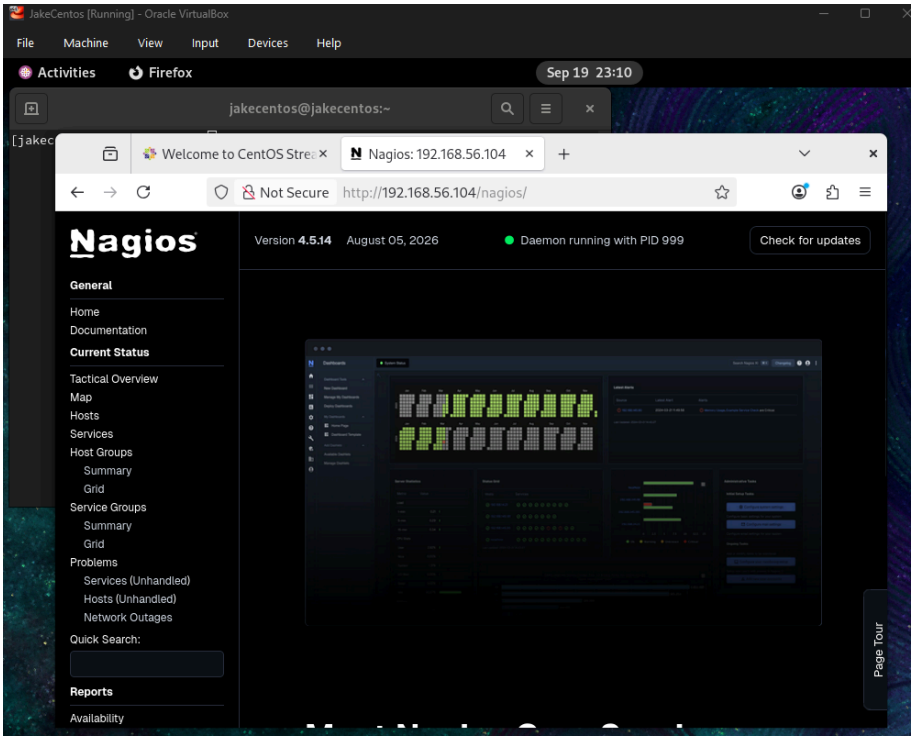
```
jakeworkstation@WS:~/JakeGestiada/HOA8$ ssh jakecentos@192.168.56.104
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sat Sep 19 23:01:29 2026 from 192.168.56.105
[jakecentos@jakecentos ~]$ sudo htpasswd /usr/local/nagios/etc/htpasswd.users nagiosadmin
[sudo] password for jakecentos:
New password:
Re-type new password:
Updating password for user nagiosadmin
[jakecentos@jakecentos ~]$
```

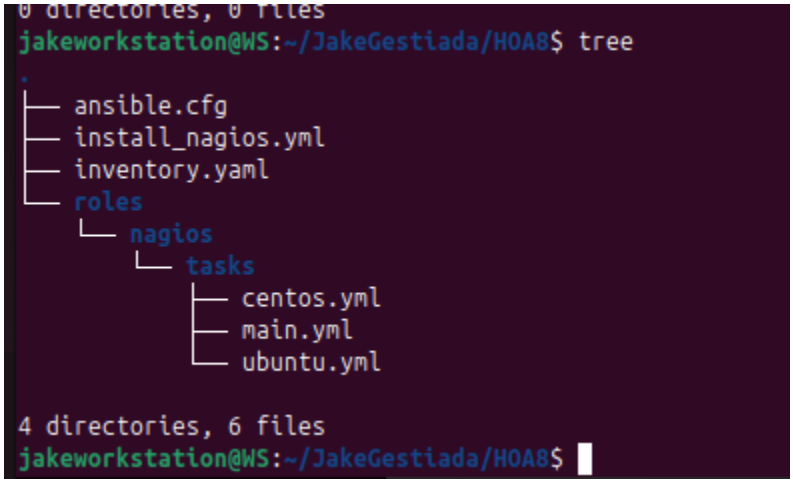
- The nagiosadmin password was successfully updated on CentOS and server1, allowing access to the Nagios web interface for verification.


Nagios: 192.168.56.105
Not Secure
http://192.168.56.105/nagios/
70%
Nagios
Version 4.5.14 August 05, 20
Save password for http://192.168.56.105?
Username
nagiosadmin
Password
Not now
Save
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Page Tour


Not Secure
http://192.168.56.105/nagios/
70%
Nagios
Version 4.5.14 August 05, 2026
Daemon running with PID 1377
Check for updates
Meet Nagios Core Services Platform
The next generation of Open Source powered monitoring with advanced dashboards, monitoring wizards, and much more!
Learn More
Newsletter Sign-Up
Copyright © 2010-2026 Nagios Core Development Team and Community Contributors. Copyright © 1999-2009 Ethan Galstad. See the THANKS file for more information on contributors.
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- The Nagios login page appeared successfully on Ubuntu Server1, and the nagiosadmin account was used to access it. After logging in, the Nagios dashboard loaded properly, confirming that the web interface was working.



- The final HOA8 folder structure was checked and confirmed to contain the required Ansible configuration, inventory, playbook, and Nagios role task files for Ubuntu and CentOS.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections

Answer the following:

What are the benefits of having an availability monitoring tool?

- Availability monitoring tools monitor whether servers and services are working well. Besides that, they assist in identifying problems before they get worse. The moment when a server or service is out, it can quickly be found. Therefore troubleshooting is made easy since it's the earlier stage that problems are pointed out. Not just troubleshooting, monitoring services also contribute to minimizing downtime and making important services still accessible to users despite issues. So, by and large, it ensures smooth and dependable server functioning.

Conclusion/learnings

- This activity, I have now a clear picture of how Nagios can be installed and configured with the help of Ansible. I got an idea of the various roles and task files to be used for Ubuntu and CentOS systems. Moreover, I got the ability to check if Nagios runs without any glitch on both the nodes. The configuration check allowed me to see that there aren't any errors or warnings. Additionally, I found out how to check and confirm the status of the Nagios web interface by accessing it via a web browser. To sum up, the activity has taught me a better way to manage server monitoring and management through automation.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly and Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **For grammar correction and debug terminal error.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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